true term
spurious intercept

**a**

Rössler $\dot{x}_1$, $n = 5000$

SNR = 61 dB

SNR = ∞

selection frequency

1.0
0.5
0.0

$x_2$ $x_3$ intercept

candidate term

**b**

Dadras $\dot{x}_1$, SNR = 49 dB

$n = 10^4$

$n = 10^5$

selection frequency

1.0
0.5
0.0

$x_1$ $x_2$ $x_2x_3$ intercept

candidate term

residual
binned mean ± 2 sd (12 bins)
zero line

**c**

Rössler $\dot{x}_1$, SNR = 61 dB

standardised residual

6
3
0
-3
-6

-20 -10 0 10

fitted value (posterior mean)

**d**

Rössler $\dot{x}_1$, SNR = ∞

standardised residual

6
3
0
-3
-6

-20 -10 0 10

fitted value (posterior mean)

$\hat{k}_i \le 0.7$
$\hat{k}_i > 0.7$ (flagged)
threshold $\hat{k}_i = 0.7$

**e**

Dadras $\dot{x}_1$, $n = 10^4$ ($t \le 100$)

Pareto $\hat{k}_i$

2
1
0

0 20 40 60 80 100

time

**f**

Dadras $\dot{x}_1$, $n = 10^5$ ($t \le 1000$)

Pareto $\hat{k}_i$

2
1
0

0 200 400 600 800 1000

time

clean trajectory (trial 1)
flagged, $\hat{k}_i > 0.7$
admitted intercept (median and range)
$1.645\ \sigma/\sqrt{n}$

**g**

Dadras, $n = 10^5$, 20 trials

$x_3$

10
0
-10

-20 -10 0 10 20

$x_1$

**h**

Dadras $\dot{x}_1$, SNR = 49 dB

intercept

0.016
0.012
0.008
0.004
0.000

1 3 11 14 30

$10^{4.6}$ $10^{4.7}$ $10^{4.8}$ $10^{4.9}$ $10^5$

$n$